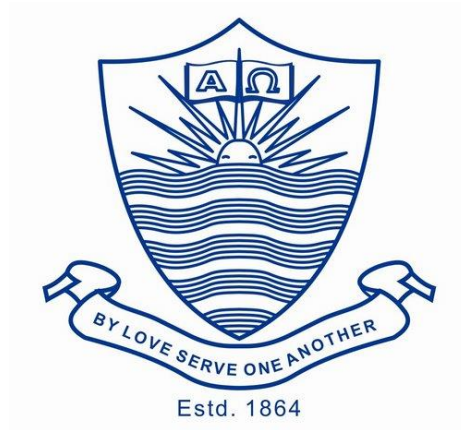


FORMAN CHRISTIAN COLLEGE (A CHARTERED UNIVERSITY)



Course Title Intro to AI

Lab 8

Title of Lab Mini-Lab: Fuzzy Logic

Group Member/s Names and Roll Number/s

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Part A – Warm-up Discussion

1.
4 hours of sleep is not strictly “low.” It can also be partly “medium” because sleep categories are gradual, not sharply divided.
 2.
Fuzzy logic is more realistic because real-life situations (like sleep and workload) are not strictly black-and-white; they exist on a spectrum.
-

Part B – Interpreting the Inputs

3.
Sleep = 4 hours belongs to:
 - Low: 0.8
 - Medium: 0.3
 4.
The values mean that 4 hours of sleep is mostly considered low (0.8), but it still has some characteristics of medium sleep (0.3). This shows overlap between categories.
 5.
Workload being High with membership 0.7 means the workload is strongly high, but not completely or perfectly high.
-

Part C – Rule Evaluation

6.
Rule:
IF sleep is LOW AND workload is HIGH
Using min operator:

$$\text{Activation} = \min(0.8, 0.7)$$

7. Calculation:

$$\text{Activation} = 0.7$$

8. Explanation:
This means the rule is activated to a degree of 0.7, indicating that the student’s stress is strongly (but not completely) very high based on the given conditions.
-

Part D – What-If Variation

9.

New values:

Sleep (Low) = 0.2

Workload (High) = 0.7

$$\text{Activation} = \min(0.2, 0.7) = 0.2$$

10.

The first student activates the rule more strongly (0.7 vs 0.2) because their sleep is much more strongly “low,” which is a key condition in the rule.

11.

This shows that fuzzy logic allows smooth, gradual changes. A small change in input (sleep) leads to a gradual change in output (stress), not a sudden jump.

Extension

12. Example Rule:

IF sleep is MEDIUM AND workload is HIGH

THEN stress is HIGH

13. Justification:

Even with moderate sleep, a high workload can still cause significant stress, though not as extreme as when sleep is very low.

Quick Reflection

14.

Fuzzy logic allows partial truth values between 0 and 1, unlike crisp logic which only allows true or false.

15.

Fuzzy logic represents degrees of truth, while probability represents uncertainty about whether an event will occur.